$$Z_{eff} \propto \frac{+\text{ive charge}}{-\text{ive charge}}$$

$$Z_{eff} = (Z - \sigma)$$

$Z_{eff}\uparrow$ → R

↓ Z_{eff} remains same.

$Li < Be < B < C < N < O$

$N^{+} > N > N^{-}$

$H^{-} < Li^{+} < Be^{+2} < Be^{+3}$ more +ive charge

$Li \approx Na \approx K$

$N^{-3} < O^{-2} < F^{-} < Na^{+} < Mg^{+2}$

→

+ive charge ↑ so $Z_{eff}\uparrow$

Radius

Vanderwall > metallic > covalent

$$\boxed{\text{Vanderwall radius} = 2 \times \text{Covalent Radius.}}$$

L → Size ↓ Li > Be > B > C > N > O

Reason :- Zeff ↑

Pull of nucleus ↑

Shrink

Li
Na
K
Rb
Cs ↓

Size ↑

Reason :-

No. of shell Increase.

Ionic radius (cation ⊕ Anion ⊖)

Cation < Parent atom < ANION

$H^+ < H < H^-$ (NEET)

More +ive less radius → $Mn^{+2} > Mn^{+4} > Mn^{+7}$

$Fe^{+2} > Fe^{+3}$

More -ive more radius $O^- < O^{-2}$

$N < N^- < N^{-3}$

Isoelectronic Species

NEET

JEE MAIN
NEET

More Z (atomic number) small size.

$$\underset{16}{S^{-2}} > \underset{17}{Cl^{-}} > \underset{19}{K^{+}} > \underset{20}{Ca^{+2}}$$

$$\underset{7}{N^{-3}} > \underset{8}{O^{-2}} > \underset{9}{F^{-}} > \underset{11}{Na^{+}} > \underset{12}{Mg^{+2}} > \underset{13}{Al^{+3}}$$

$$Na^{+} > Li^{+} > Mg^{+2} > Be^{+2}$$

माली मॉगे Berrylium.

(1) Li > Be > B > C > N > O > F (along Pd)

(2) Li < Na < K < Rb < Cs (Along a group ⇒ size ↑)

** (3) Sc > Ti > V > Cr > Mn > Fe ≈ Co ≈ Ni < Cu < Zn

(4) $\overset{3d}{Sc} < \overset{4d}{Y} < \overset{5d}{La}$ (along group III B)

** (5) Ti < Zr ≈ Hf (Lanthanoid contraction)

** (6) B < Al ≈ Ga < In < Tl

$\boxed{B < Ga < Al < In < Tl}$

Transition contraction

@bohring_bot

S- block

$Li < Mg$

$Li^+ > Mg^{+2}$

Li   Be

Na   Mg

K   Ca

Rb   Sr

Cs   Ba

Inert gas

↳ largest in thier period.

↳ vanderwall Radius is counted

$\boxed{Ne > O}$

$\boxed{He} > H$

Ionisation potential $\propto \dfrac{1}{\text{size}}$

L increase ↑ R

decrease

$Li < Be < B < C < N < O < F$ XX

✓ $Li < B < Be < C < O < N < F$ Imp

$Li > Na > K > Rb > Cs$

$Be > B$

s-electron p-electron

(p^3) $N > O$

half filled

Grp-13

1 B

2 Al

3 Ga

4 In

5 Tl

$\boxed{1\ 5\ 3\ 2\ 4}$

B > Tl > Ga > Al > In

I.E. →

$\boxed{**}$ W-shape graph बनेगा —

(I.E. and atomic Number)

Grp-14

C > Si > Ge > Sn > Pb

C > Si > Ge > Pb > Sn

(I.E.)

Q.

IE_1	IE_2	IE_3	IE_4	IE_5
7.1	14.3	34.5	46.8	162.2

Largest Jump

No. of valence electron = Largest Jump में छोटा वाला

$4 \longrightarrow 5$

$n = 4$ (valence e^-)

$\boxed{\text{Always}}$

↳ $IE_1 < IE_2 < IE_3$

↳ IE is always +ive (Endothermic)

ΔH_{eg}

Electron Affinity

Increase ↑ →

decrease ↓

$$O + e^- \longrightarrow O^-$$

$$\Delta H_{eg_1} = -ive \ \text{(Exothermic)}$$

$$O^- + e^- \longrightarrow O^{-2}$$

$$\Delta H_{eg_2} = +ive \ \text{(Endothermic)}$$

Always.

$$O \xrightarrow[\boxed{Exo}]{} O^- \xrightarrow[\boxed{Endo}]{} O^{-2}$$

2nd pd

$Li < \boxed{Be} < B < C < \boxed{N} < O < F < \boxed{Ne}$

$Ne < Be < N < B < Li < C < O < F$

★★

3rd pd > 2nd pd

max

$Cl > F > Br > I$

$S > O > P > N$

3rd pd 2nd pd.

$Si > C > P > N$

stable

F
Cl
Br
I

Electronegativity $\propto$ % s character

$F > Cl > Br > I$

B

Al

Ga

L — Increase → R

Decrease ↓

$Mn^{+2} < Mn^{+4} < Mn^{+7}$

$O^{-2} < O^{-1} < O < O^{+1} < O^{+2}$

$Fe < Fe^{+2} < Fe^{+3}$

$$EN \propto \frac{+ive}{-ive}$$

$\overset{3.25}{C \equiv C} > \overset{2.75}{C = C} > \overset{2.5}{C - C}$

sp sp2 sp3

$\boxed{**}$ Ga > Al (Exception) $\boxed{**}$

Nature of Oxide (Metal oxide Basic) {MOB}

L metallic character↓ R

→ acidic character↑

Basic character↓

↓

Basic character↑

metallic character↑

$SO_2 > P_2O_5 < Cl_2O_7$

(acidic)

$B_2O_3 < CO_2 < N_2O_5$

$NO_2 > ZnO > K_2O$

K metallic

N non metal

IMP

Amphoteric

BeO

SnO / SnO_2

Al_2O_3

ZnO

PbO / PbO_2

Sb_2O_3

Neutral → CO, H_2O, NO, N_2O

$\boxed{\text{Exception}}$ $\overset{+5}{H_3PO_4} < \overset{+3}{H_3PO_3} < \overset{+1}{H_3PO_2}$

Acidic strength of oxy acid $\propto$ Oxidation state

$\overset{+7}{HClO_4} > \overset{+5}{HClO_3} > \overset{+3}{HClO_2} > \overset{+1}{HClO}$

$\overset{+6}{H_2SO_4} > \overset{+4}{H_2SO_3}$

$Sb_2O_5 > Sb_2O_3$

$\overset{+6}{SO_3} > \overset{+4}{SO_2}$

$\overset{+5}{HNO_3} > \overset{+3}{HNO_2}$ (Nitrous acid)

$\overset{+5}{N_2O_5} > \overset{+3}{N_2O_3}$

Acidic strength $\propto$ Electronegativity of central atom.
(along a period)

$HNO_3 > H_2CO_3 > H_3BO_3$ $\boxed{EN}$ $N > C > B$

$HClO_4 > HBrO_4 > HIO_4$

$N_2O_3 > P_2O_3 > As_2O_3 > Sb_2O_3$

$H_2SO_3 > H_2SeO_3 > H_2TeO_3$

Acidic strength (Bond Energy कम)

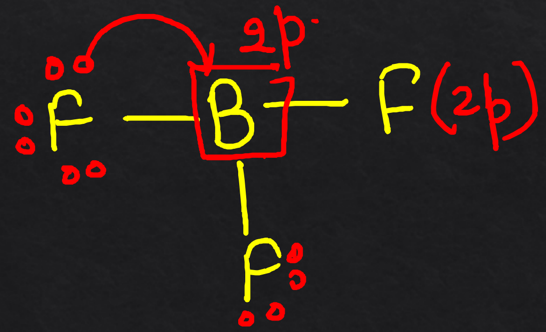
$HF < HCl < HBr < HI$ (Bond length ज्यादा)

$H_2O < H_2S < H_2Se < H_2Te$

$\boxed{BF_3} < BCl_3 < BBr_3 < BI_3$

BF_3 → very less acidic

↳ Back bonding



Bond angle $\propto \frac{1}{\text{lone pair}} \propto$ % s character

$\propto$ EN of central atom

$AsH_3 < PH_3 < NH_3$ [E.N.]

$H_2Se < H_2S < H_2O$

$NF_3 < NCl_3$ [Cl large ⇒ repulsion]

$\ddot{N}O_2^- < \dot{N}O_2 < NO_2^+$

1 lone pair sp^2

1 free electron sp^2

sp (B.A. = 180°)

sp^3 CH_4 < C_2H_4 (sp^2) < C_2H_2 (sp) H−C≡C−H

★★ NF_3 < NH_3 < NCl_3

Size : ($H^- > F^-$)

PF_3 < PH_3 < PCl_3 < PBr_3 < PI_3

sp^3	sp^3	sp^3	sp^2
$H_2\ddot{\underset{..}{O}}$ <	$\ddot{N}H_3$ <	CH_4 <	BH_3
(104°5')	(107°)	(109°28')	(120°)

SF_6 < H_2O < NH_3 < CH_4 < CO_2 (liniar) [B.A.=180°]

sp^3d^2 sp^3 sp^3 sp^3 sp

OF_2 < OH_2 < Cl_2O

Thermal stability $\propto$ Lattice Energy $\propto \dfrac{\text{charge}}{\text{size}}$ $\propto$ Ionic charact

(i) Size ⇒ Size ↑ Bond length ↑ Bond Energy ↓

Thermal stability ↓

(ii) monoatomic anion ⇒ Ionic ⇒ Lattice Energy

(H^-, O^-, Cl^-, F^-)

Small size की Thermal stability ↑

(iii) Polyatomic anion ⇒ covalent character

CO_3^{-2}, SO_4^{-2}, OH^- ⇒ Fajan's Rule.

NO_3^-

large की thermal stability ज्यादा।

most covalent

$LiOH < NaOH < KOH < RbOH < CsOH$

large size

$SbH_3 < AsH_3 < PH_3 < NH_3$

large size

$HI < HBr < HCl < HF$

most covalent

$BeCO_3 < MgCO_3 < CaCO_3 < BaCO_3$

$CsH < RbH < KH < NaH < LiH$

small size

High Lattice energy

L.E.

$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3$

covalent

L.E. ← $BaH_2 < SrH_2 < CaH_2 < MgH_2 < BeH_2$

covalnt ← $BeSO_4 < MgSO_4 < CaSO_4$

$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2$

$CCl_4 < BCl_3 < BeCl_2 < LiCl$

most covalent

so Thermal stability कम

Solubility

$BaSO_4$ → Insoluble (Dye) white
radioopaque. → X-ray

$BaCO_3 < CaCO_3 < MgCO_3 < BeCO_3$

NCERT

$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Ba(OH)_2$

Soluble

LE < HE

$BaSO_4 < SrSO_4 < CaSO_4 < MgSO_4$

LiF NaF KF RbF CsF

Covalent

$\boxed{AgI} < AgBr < AgCl < AgF$

IA

Salt

$\boxed{\text{Ionic}}$

Lattice Energy

more solubility

$LiF < NaF < KF < RbF$

$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3$

$LiOH < NaOH < KOH < RbOH$

$LiF < LiCl < LiBr < LiI$

Insoluble b/c very high Lattice Energy.

@bohring_bot

Ionic character

Fajan's Rule → Small cation
Large anion
more charge
Pseudo inert gas configuration

More covalent

covalent

Small cation

$BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2$

$BCl_3 < AlCl_3 < GaCl_3$

$\overset{+4}{V}Cl_4 < \overset{+3}{V}Cl_3 < \overset{+2}{V}Cl_2$

Charge more

$LiBr < NaBr < KBr < RbBr < CsBr$

Small cation

$HI < HBr < HCl < HF$

large anion → covalent

$\overset{+3}{AlF_3} < \overset{+2}{MgF_2} < \overset{+1}{NaF}$

size $N^{-3} > O^{-2} > F^{-}$

$AlN < Al_2O_3 < AlF_3$

large anion

$\underline{CuCN} < Ag\underline{CN}$

Cu

$\boxed{Ag}$

$AgCl < \underline{KCl}$

Oxidising power $\propto$ oxidation state

$I_2 < Br_2 < Cl_2 < \boxed{F_2}$ → strongest oxidising agent ↳ due to high hydration

$\overset{+6}{MnO_4^{-2}} < \overset{+7}{MnO_4^{-}}$

$\overset{+6}{Cr_2O_7^{-2}} < \overset{+7}{MnO_4^{-}}$

$GeCl_4 < SnCl_4 < \boxed{PbCl_4}$

$\boxed{Pb^{+2} > Pb^{+4}}$ due to inert pair effect

NEET

Boiling point $\propto$ Vanderwaal force $\propto$ Molecular weight

$HCl < HBr < HI < \boxed{HF}$ ← H-bonding

NEET → $PH_3 < AsH_3 < \boxed{NH_3} < SbH_3$ → H-bonding

$H_2S < H_2Se < H_2O$ → (weaker than HF)

He < Ne < Ar < Kr

No H bonding $\boxed{CH_4}$ < NH_3 < HF < H_2O

4 2 4 H-bond

$\boxed{\text{weak}}$

H_2O < D_2O

H_2 < Cl_2 < Br_2

Extent of hydrolysis $\propto$ covalent character

No vaccant orbital

So NO HYDROLYSIS

$\overset{+4}{CCl_4} < \overset{+2}{MgCl_2} < \overset{+3}{AlCl_3} < \overset{+4}{SiCl_4} < \overset{+5}{PCl_5}$

$BiCl_3 < SbCl_3 < AsCl_3 < PCl_3 < NCl_3$

Bond strength

Size ↑
Bond length ↑
Bond Energy ↓

$HI < HBr < HCl < HF$

$N-N < N=N < N\equiv N$

Bond order.

	1	1.5	2	2.5	3.0
	O_2^{-2} <	O_2^{-} <	O_2 <	O_2^{+1} <	O_2^{+2}

Bond order ↑ Bond length ↓

Bond Energy ↑

V.Imp → $I_2 < F_2 < Br_2 < Cl_2$ =

(2nd pd) $\ddot{O}-\ddot{O} < \ddot{S}-\ddot{S}$ (3rd pd)

$F-F < O-O < N-N < C-C < H-H$

3 l.p. 2 l.p. 1 l.p. 0 l.p. 0 l.p.

$\leftarrow \quad \rightarrow$

$:\ddot{F}-\ddot{F}:$ repulsion

Bond Energy

धन्यवाद

Telegram → Dr. Anurag pandey

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